

2. Coordinate geometry in the (x, y) plane

What students need to learn:

Equation of a straight line, including the forms $y - y_1 = m(x - x_1)$ and $ax + by + c = 0$.

To include:

- (i) the equation of a line through two given points
- (ii) the equation of a line parallel (or perpendicular) to a given line through a given point. For example, the line perpendicular to the line $3x + 4y = 18$ through the point $(2, 3)$ has equation $y - 3 = \frac{4}{3}(x - 2)$.

Conditions for two straight lines to be parallel or perpendicular to each other.

Coordinate geometry of the circle using the equation of a circle in the form $(x - a)^2 + (y - b)^2 = r^2$ and including use of the following circle properties:

Students should be able to find the radius and the coordinates of the centre of the circle given the equation of the circle, and vice versa.

- (i) the angle in a semicircle is a right angle;
 - (ii) the perpendicular from the centre to a chord bisects the chord;
 - (iii) the perpendicularity of radius and tangent.
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